AMD NVIDIA Pair Trading Preliminary Validation Results

QF600 Asset Pricing

The preliminary validation does not support a robust full-sample mean-reversion relationship between AMD and NVIDIA. Although their daily returns display moderate positive co-movement, the cointegration and residual-stationarity tests do not provide evidence that the estimated spread is stable over the full sample.

Key Results

Analysis input	**Value**
Assets	AMD and NVDA
Price series	Daily adjusted closing prices
Sample period	2 January 2015 to 11 September 2026
Common trading days	2,938

Measure	**Result**	**Reading**
Daily return correlation	0.570	Moderate positive co-movement
Engle-Granger statistic	-1.603	Diagnostic statistic
Engle-Granger p-value	0.720	No evidence against no cointegration
ADF statistic for residual spread	-1.602	Diagnostic statistic
ADF p-value for residual spread	0.482	No evidence of a stationary residual spread
Static regression intercept	1.732	Log-price regression coefficient
Static hedge ratio	0.783	Full-sample log-price coefficient

Note. A p-value below 0.05 is a common benchmark for rejecting the relevant null hypothesis.

Rolling Diagnostics

The full-sample hedge ratio is **positive at 0.783,** whereas the most recent 60-day rolling estimates are **negative,** ranging from **-0.213 to -0.465.** This change in sign is a practical indication that the estimated relationship varies materially across time. The hedge ratio should therefore be treated as a diagnostic regression coefficient rather than as a fixed trading weight.

Recent rolling z-scores range from -1.216 to -0.272. None reaches the commonly used illustrative thresholds of +2 or -2. Accordingly, the latest observations do not indicate an extreme spread deviation under those reference levels. These thresholds have not yet been selected as formal trading rules.

Date	Rolling alpha	Hedge ratio	Log spread	Z-score
28 Aug 2026	7.363	-0.213	-0.072	-1.216
31 Aug 2026	7.607	-0.260	-0.052	-1.071
1 Sep 2026	7.852	-0.306	-0.076	-1.168
2 Sep 2026	8.048	-0.343	-0.070	-1.094
3 Sep 2026	8.259	-0.382	-0.061	-1.013
4 Sep 2026	8.479	-0.423	-0.008	-0.704
8 Sep 2026	8.704	-0.465	0.042	-0.416
9 Sep 2026	8.626	-0.450	0.066	-0.272
10 Sep 2026	8.476	-0.422	0.019	-0.500
11 Sep 2026	8.483	-0.424	0.045	-0.340

Analysis and Interpretation

The return correlation of **0.570** indicates that AMD and NVIDIA often move in the same direction on a daily basis. This is not sufficient for pairs trading. A mean-reversion strategy needs the relative log-price relationship, rather than simply the two returns, to be stable over time.

The full-sample Engle-Granger p-value of **0.720** is well above 0.05. The ADF p-value of **0.482** also fails to reject non-stationarity in the residual spread. Taken together, **these tests do not support a claim** that the AMD-NVIDIA spread is cointegrated or reliably mean-reverting across the full sample.

Overall, the preliminary analysis is useful because it identifies an important limitation before a backtest is constructed: AMD and NVIDIA do not show robust full-sample cointegration. Any subsequent pairs-trading analysis should make this limitation explicit and should not claim that a rolling z-score alone establishes a valid mean-reversion opportunity.